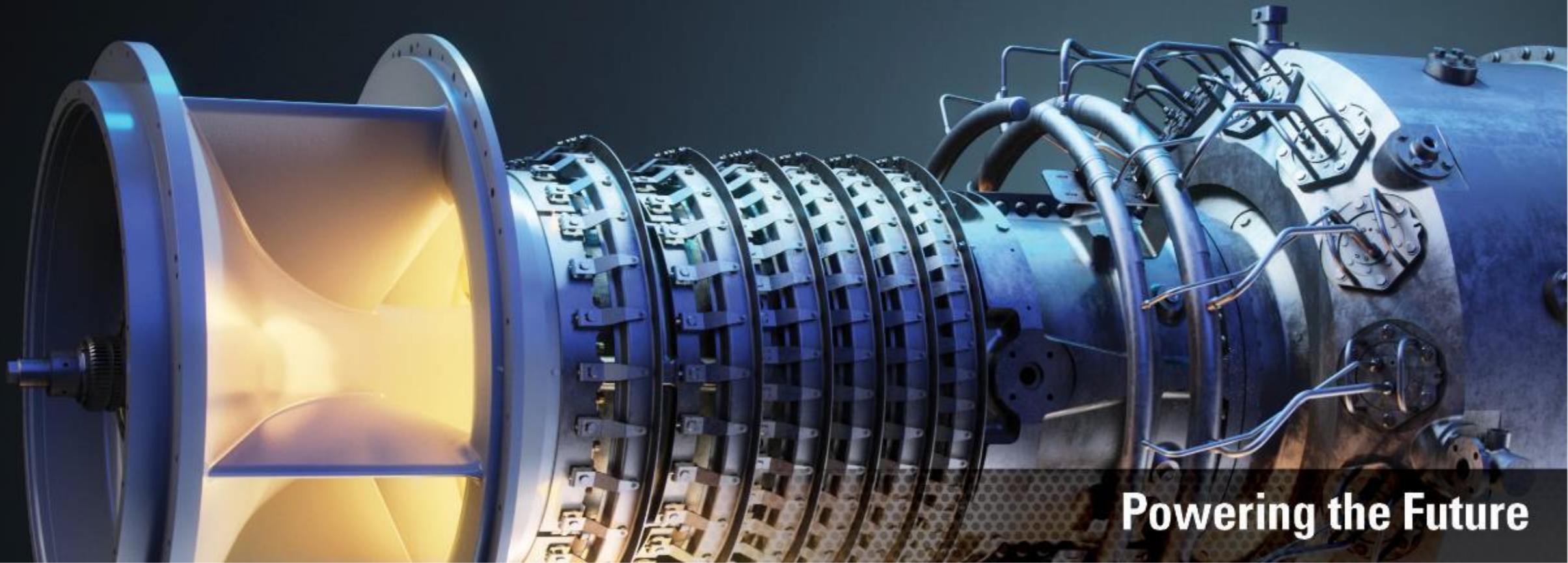


Compressor Rotor Joints and Interference Fit Study



Powering the Future

Overview

- Background
- Project Scope and Goals
- Initial Fixed-Free ANSYS Model & Results
- Manufacturing Process
- Fixed-Free Modal Impact Testing
- Refined Free-Free ANSYS Model & Results
- Free-Free Modal Impact Testing Results
- Cost
- Discussion/Conclusion

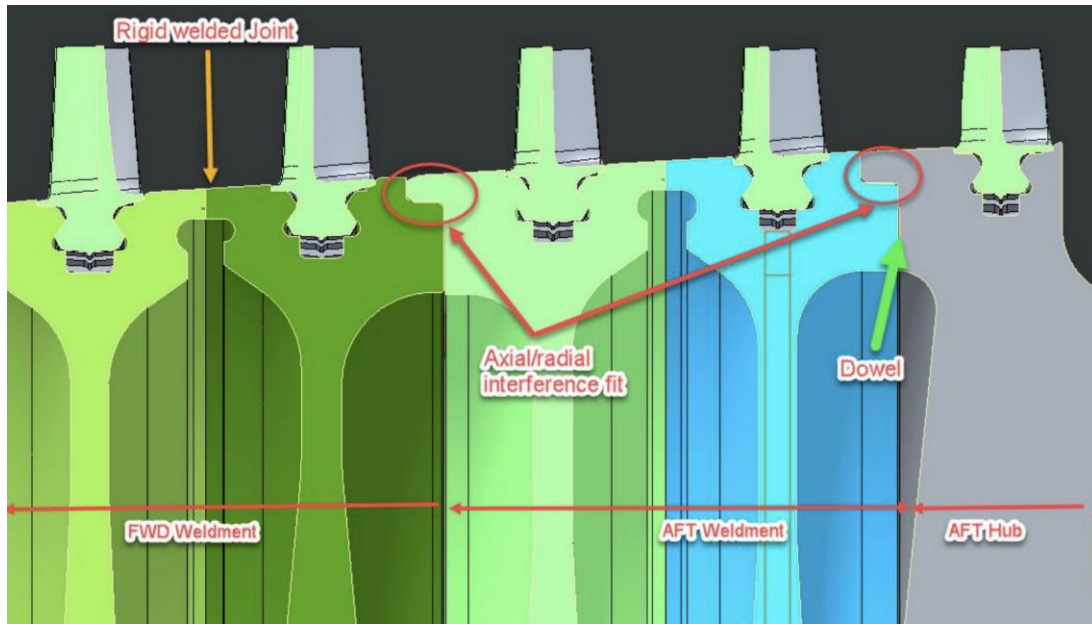


Senior Project Expo

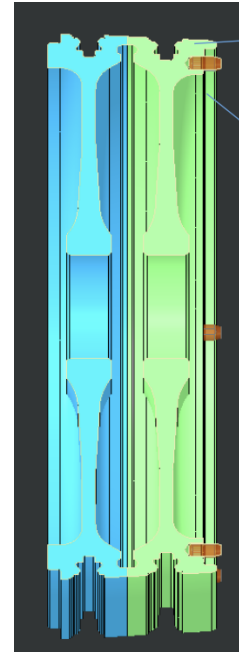
AP

Background

- FWD weldment of the compressor turbine assembly is connected to the AFT weldment through dowel pin interference fits and radial and axial pilot ledge interference fits
- Solar Turbines expressed concern in the radial “shifting” between weldments after standard operation
- The group’s task was to study the effects the interference fits had on the stiffness of this joint



Cross sectional view of FWD and AFT weldments



FWD
weldment
plates with
dowel pins



Manufactured compressor
plates mated together

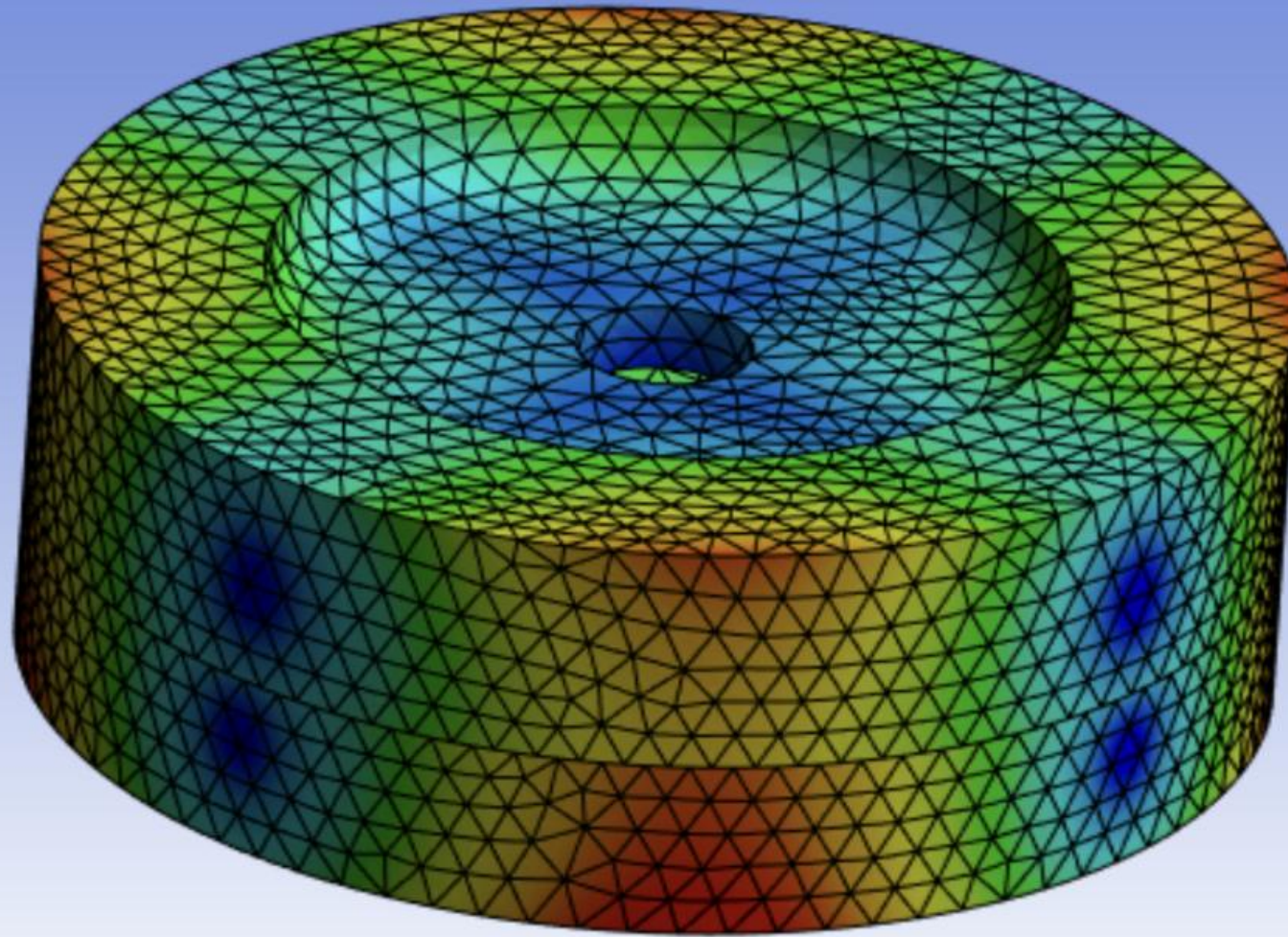
IJ

Project Scope and Goals

- Create an ANSYS model that studies the stiffness of the assembly from different dowel pin interference fits, pilot interference fits, and interface contact areas
 - Neglect axial pilot interference fit due to large bolt preload values
- Manufacture 3 scaled-down models, those being a welded assembly and two configurations of our choice
 - Low interference configuration (0.001in – 0.002in) with short pilot ledge overlap
 - High interference configuration (0.002in – 0.004in) with long pilot ledge overlap
- Testing models via modal impact hammer to experientially validate our Ansys modeling tool

IG

Initial ANSYS Model

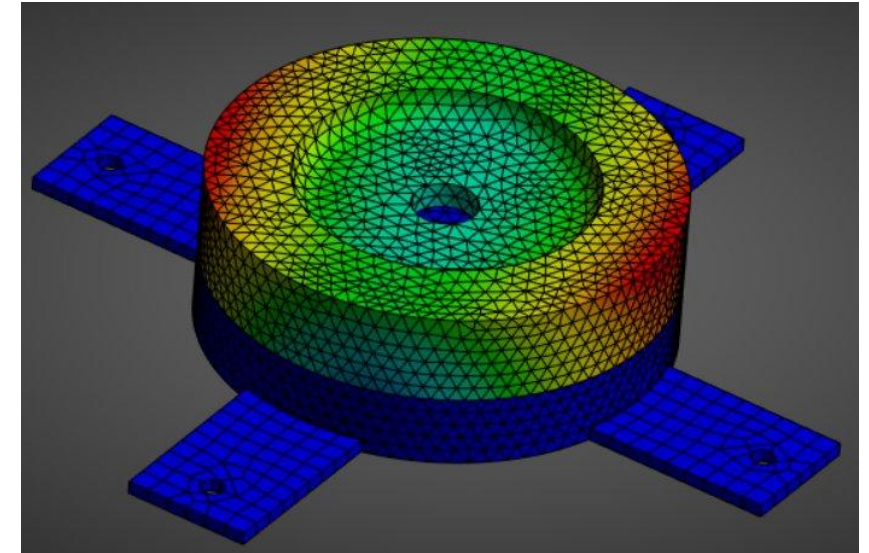
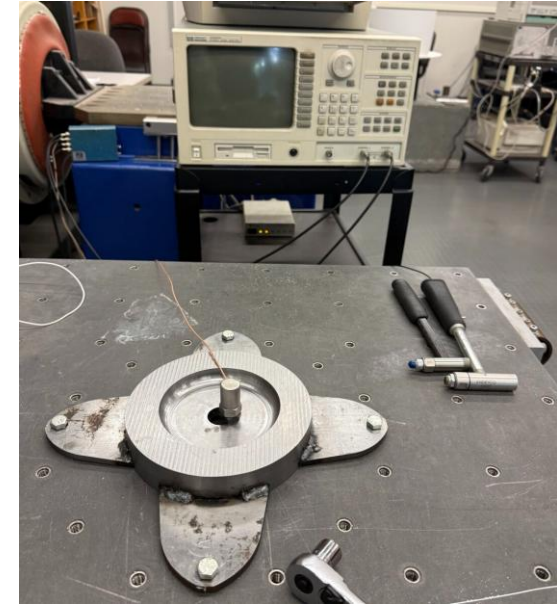


IG

Initial Fixed-Free ANSYS Model

Original Plan

- Apply a fixed boundary condition on one side of plates
- Iterate interferences and pilot ledge lengths
- Utilize the shake table to complete our testing



IG

ANSYS Model Setup

Geometry included:

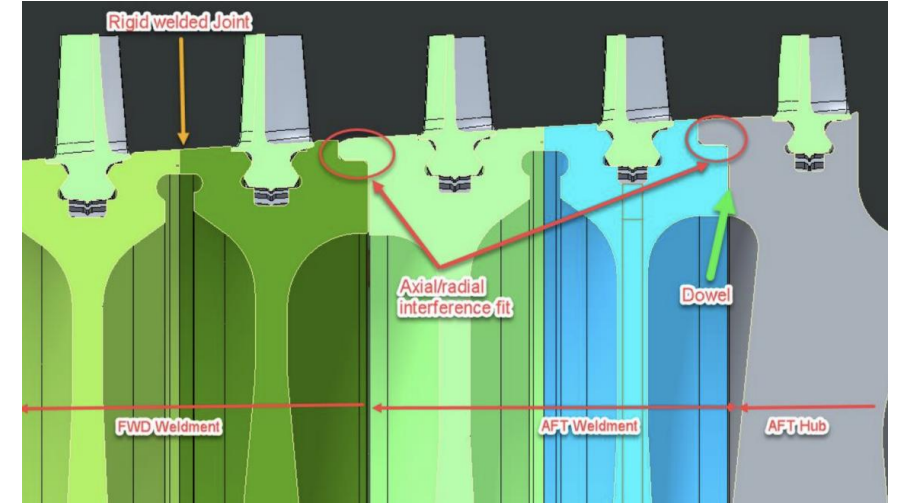
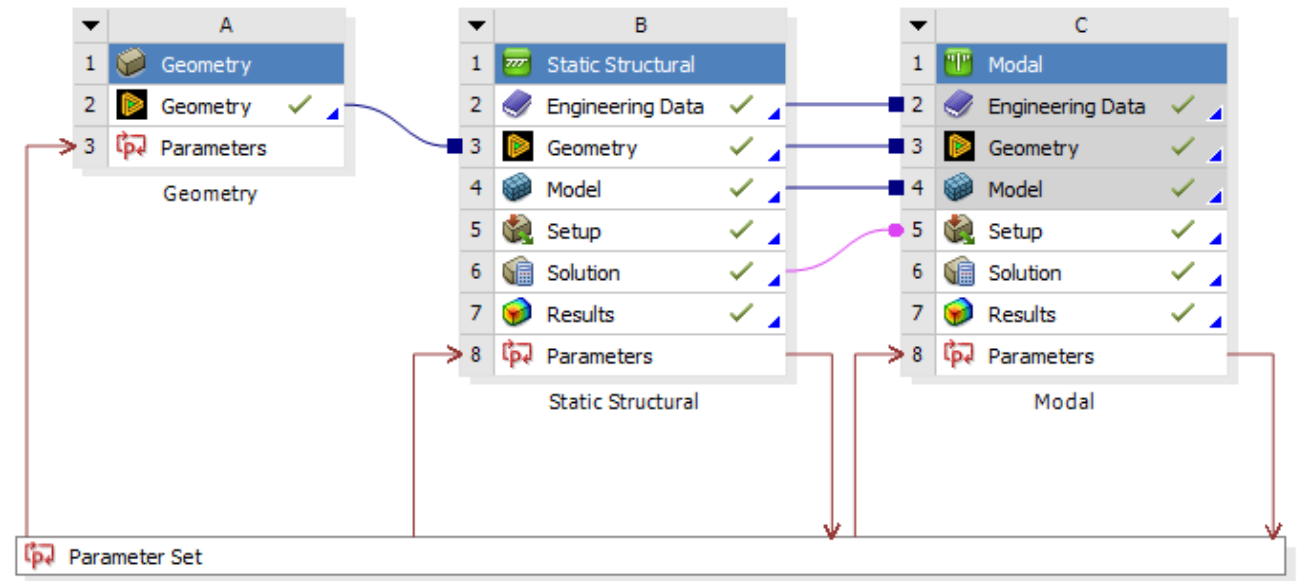
- Dowel pins
- Front half of the forward weldment and back half of the aft weldment

Iteration variables:

- Dowel pin interference fit
- Radial pilot interference fit
- Interface pilot ledge contact area

Analyses performed:

- Contact stress from interference fits
- Modal analysis



IG

ANSYS Model Setup

#1	#2	#3
Pilot Length	Female Radius	Hole Radius
0.146	3.1494	0.0937
0.15	3.1493	0.09365
0.175	3.1492	0.0936
0.2	3.1491	0.09355
0.225	3.149	0.0935
0.25	3.1489	0.09345
0.275	3.1488	0.0934
0.3	3.1487	0.09335
0.325	3.1486	0.0933
0.35	3.1485	0.09325
0.375	3.1484	
0.4	3.1483	
	3.1482	
	3.1481	
	3.148	
	3.1479	
	3.1478	
	3.14775	

Locational Interference Limits and Fits For Cylindrical Parts [(ANSI B4.1-1967,R1987)]

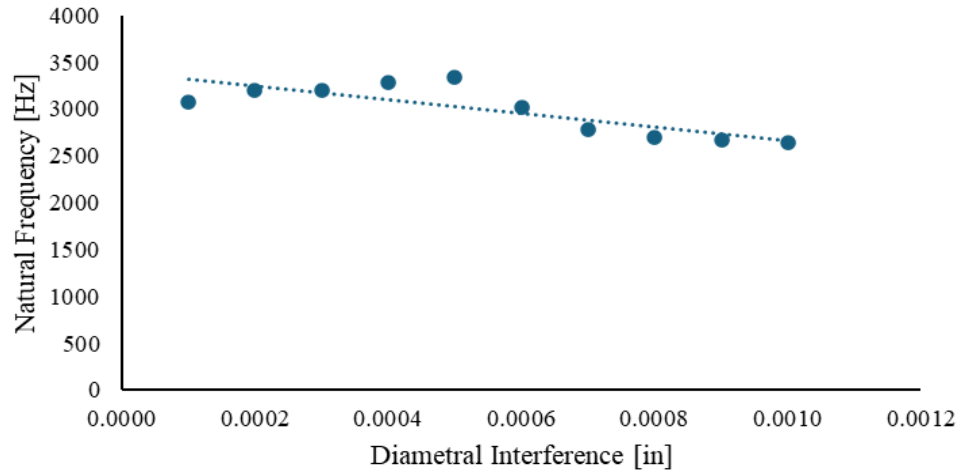
All limits shown in chart below are in thousandths of an inches. Symbols H7,p6, etc. are Shaft and Hole designations used in American-British-Canadian System (ABC). Limits for Shaft and Hole are applied algebraically to the Nominal (basic) size to obtain the limits of the size for the parts. All data given in bold in the chart below are in accordance with ABC agreements. The values given under the "Clearance Limits Column" represent min and max amounts of clearance resulting from application of standard tolerance limits.

		VALUES SHOWN BELOW ARE IN THOUSANDTHS OF AN INCHES								
Nominal (Basic) Size Renges (Inches)		Class LN1			Class LN2			Class LN3		
		Interference Limits	Standard Tolerance Limits		Interference Limits	Standard Tolerance Limits		Interference Limits	Standard Tolerance Limits	
			Hole H6	Shaft n5		Hole H7	Shaft p6		Hole H7	Shaft r6
Over	To									
0	0.12	0 0.45	+0.25 0	+0.45 +0.25	0 0.65	+0.4 0	+0.65 +0.4	0.1 0.75	+0.4 0	+0.75 +0.5
0.12	0.24	0 0.5	+0.3 0	+0.5 +0.3	0 0.8	+0.5 0	+0.8 +0.5	0.1 0.9	+0.5 0	+0.9 +0.6
0.24	0.40	0 0.65	+0.4 0	+0.65 +0.4	0 1.0	+0.6 0	+1.0 +0.6	0.2 1.2	+0.6 0	+1.2 +0.8
0.40	0.71	0 0.8	+0.4 0	+0.8 +0.4	0 1.1	+0.7 0	+1.1 +0.7	0.3 1.4	+0.7 0	+1.4 +1.0
0.71	1.19	0 1.0	+0.5 0	+1.0 +0.5	0 1.3	+0.8 0	+1.3 +0.8	0.4 1.7	+0.8 0	+1.7 +1.2
1.19	1.97	0 1.1	+0.6 0	+1.1 +0.6	0 1.6	+1.0 0	+1.6 +1.0	0.4 2.0	+1.0 0	+2.0 +1.4
1.97	3.15	0.1 1.3	+0.7 0	+1.3 +0.8	0.2 2.1	+1.2 0	+2.1 +1.4	0.4 2.3	+1.2 0	+3.2 +1.6
3.15	4.73	0.1 1.6	+0.9 0	+1.6 +1.0	0.2 2.5	+1.4 0	+2.5 +1.6	0.6 2.9	+1.4 0	+2.9 +2.0
4.73	7.09	0.2 1.9	+1.0 0	+1.9 +1.2	0.2 2.8	+1.6 0	+2.8 +1.8	0.9 3.5	+1.6 0	+3.5 +2.5

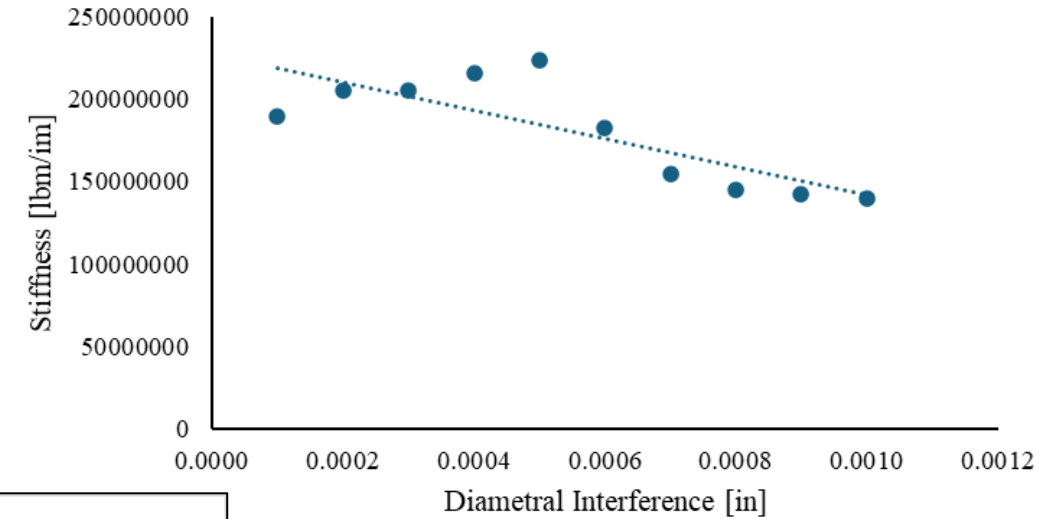
IG

ANSYS Model Results (Dowel Pin)

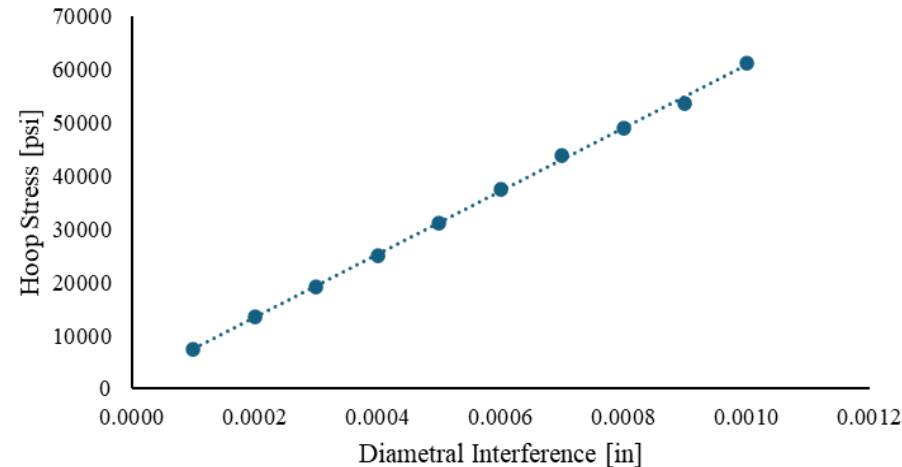
Natural Frequency vs. Interference



Assembly Stiffness vs. Interference

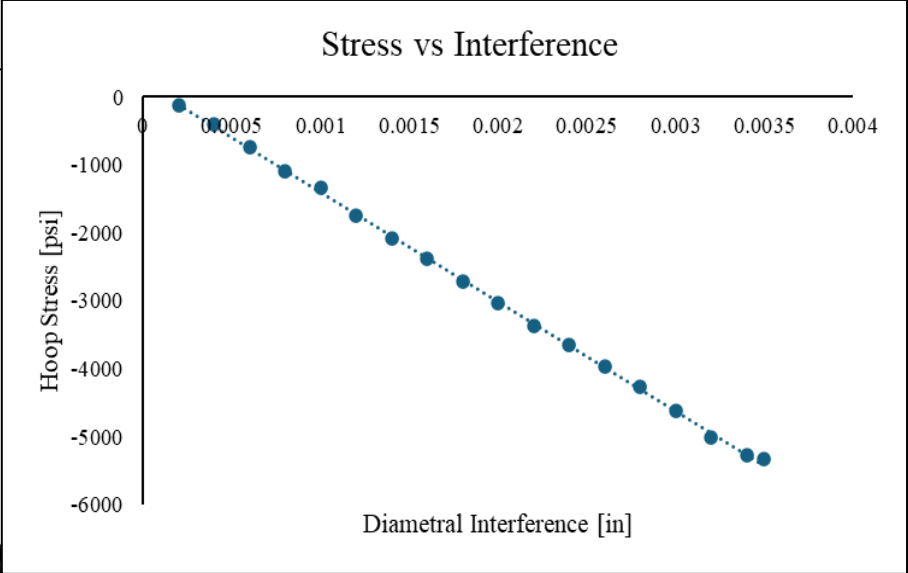
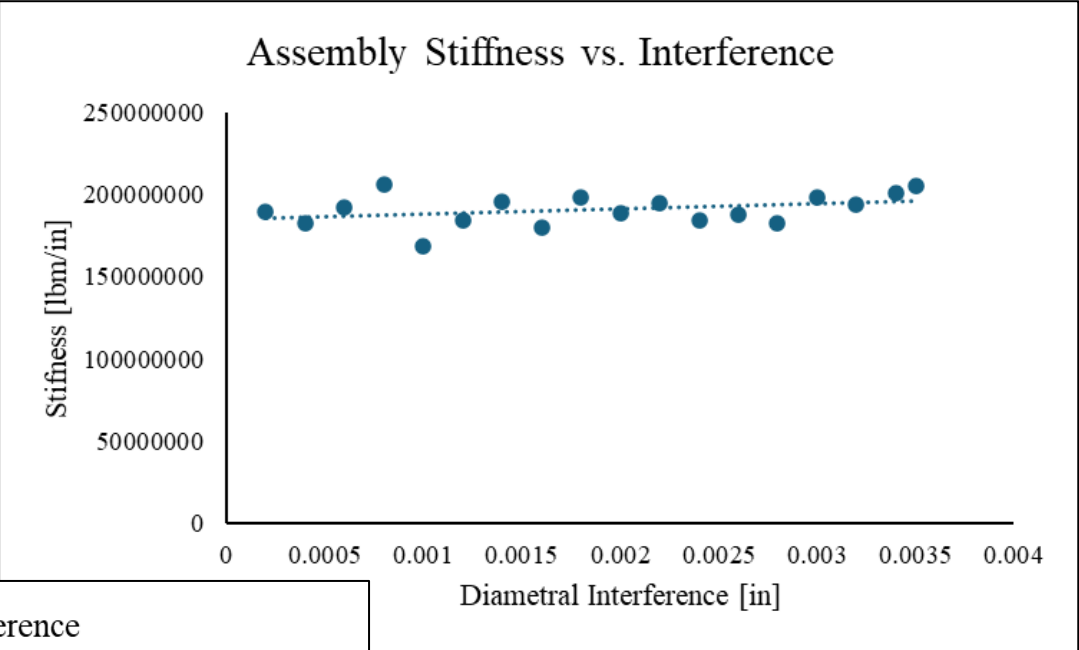
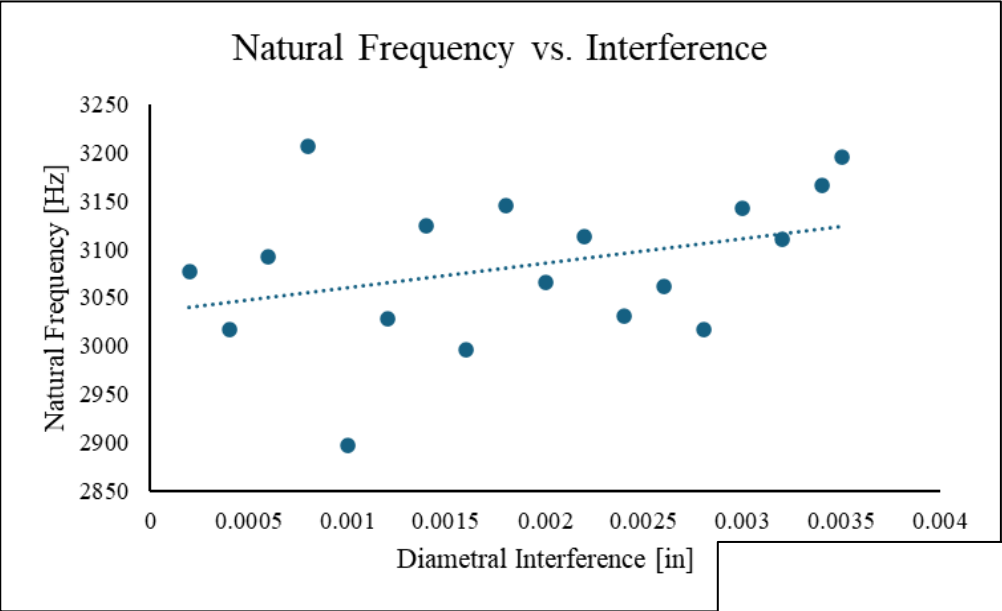


Stress vs. Interference



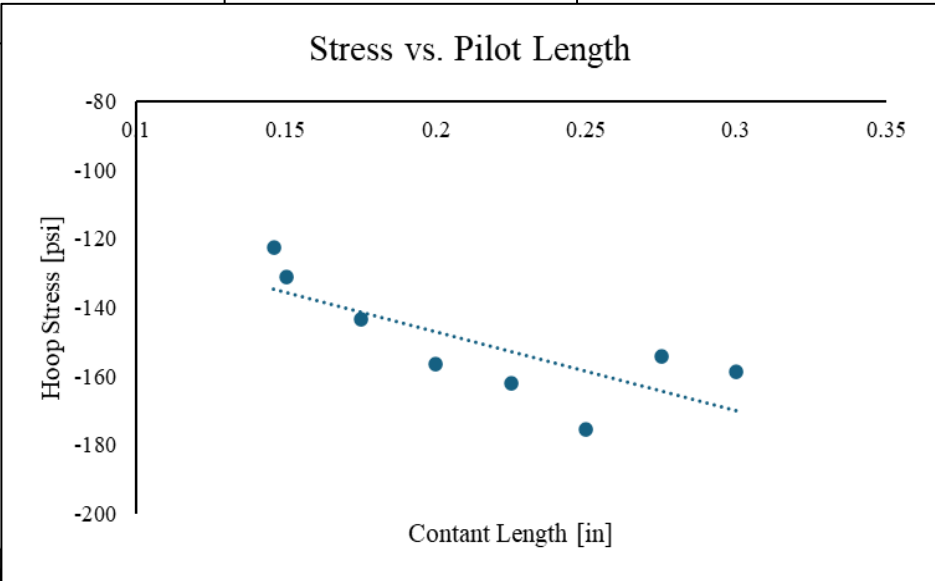
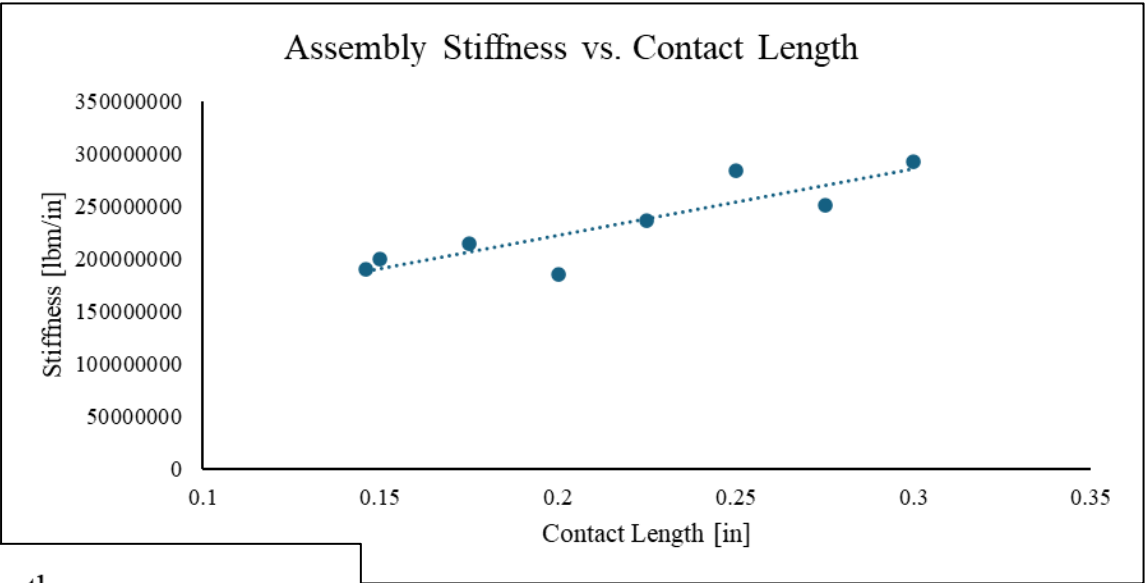
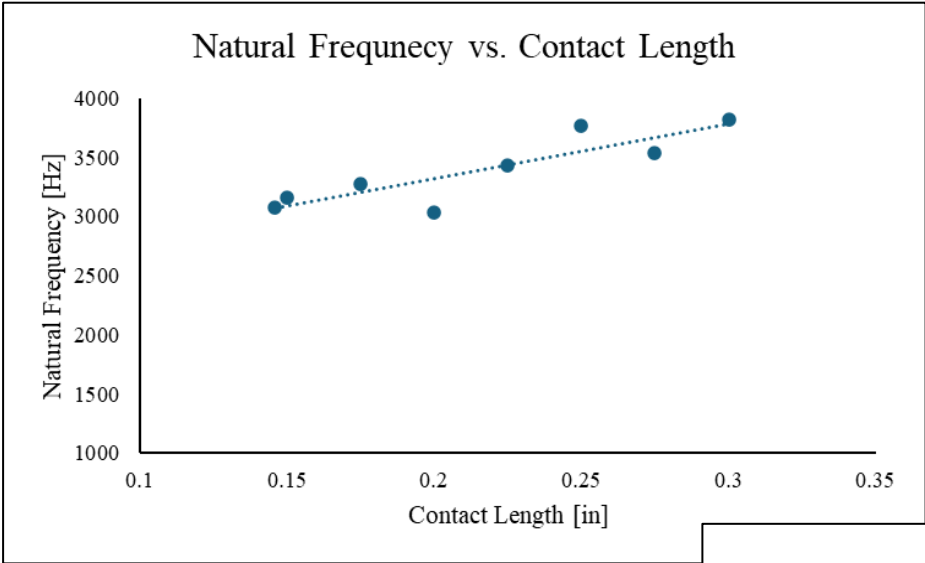
IG

ANSYS Model Results (Pilot)



IG

ANSYS Model Results (Pilot Contact Length)



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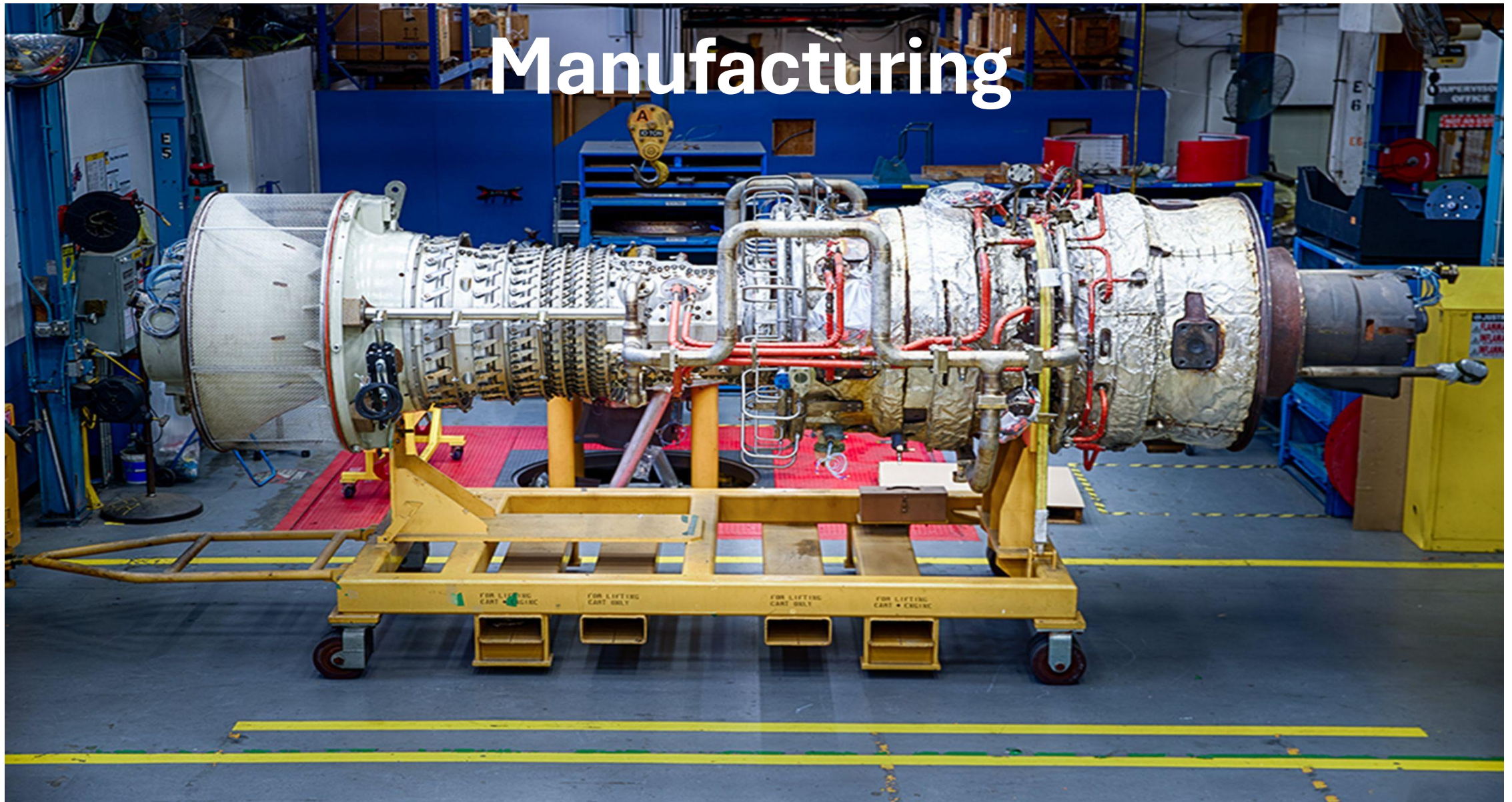
ANSYS Model 1 Key Results

Takeaways

- Iterating dowel pins did NOT significantly affect stiffness
 - Too small relative to plates
- Iterating radial pilot fit did NOT significantly affect stiffness
 - Stayed within locational interference fit range
- Iterating the pilot contact surface DID significantly affect stiffness!
 - More contact area restricted the movements of the mode shape

IG

Manufacturing



11

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Manufacturing - Hardware



Soft Jaws



TM-1 CNC



Drill press



Hydraulic Press

IJ

Manufacturing - Hardware



2 flute-ball Endmills



4 flute-
Endmills

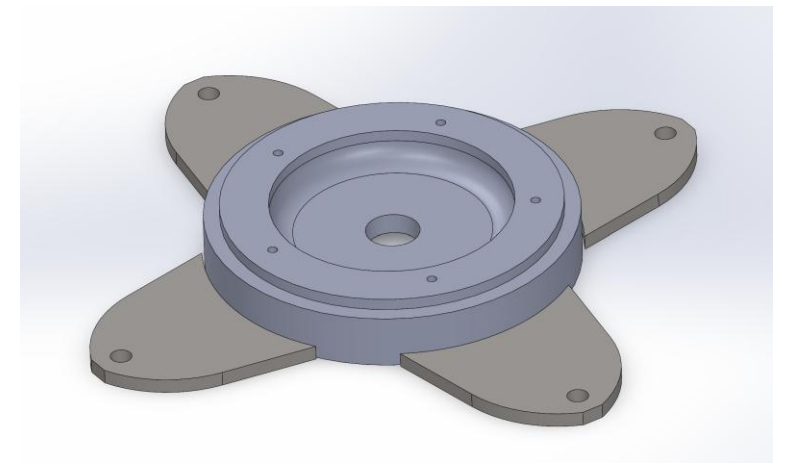


4-Reamers

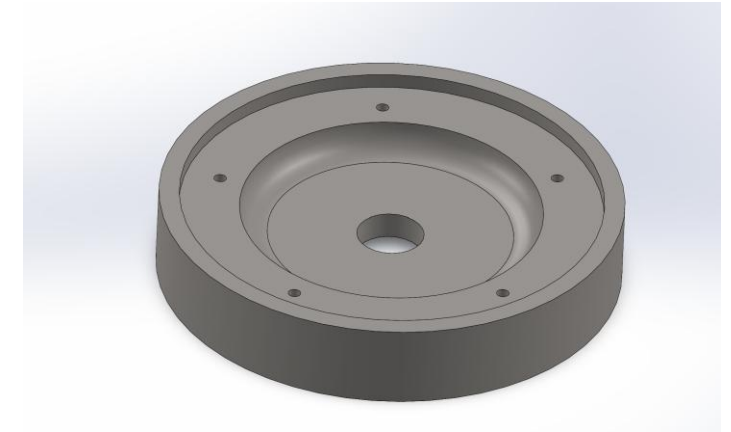
Manufacturing - CNC

Female/Male Plates

- Material: 1018 steel round stock
- CNC mill
 - 1st operation:
 - Work-holding: custom soft jaws
 - Face off top of stock, reduce diameter to 6.9"
 - 1" Center hole
 - Machine top pocket
 - 2nd operation:
 - Work-holding: custom soft jaws
 - Reduce thickness to 1.25"
 - Dowel pin holes, pilot fit ledges/pockets
- For female plate:
 - Waterjet tabs, drill press mounting holes
 - 3D print welding jig to ensure correct tab positioning on plate



Male Plate



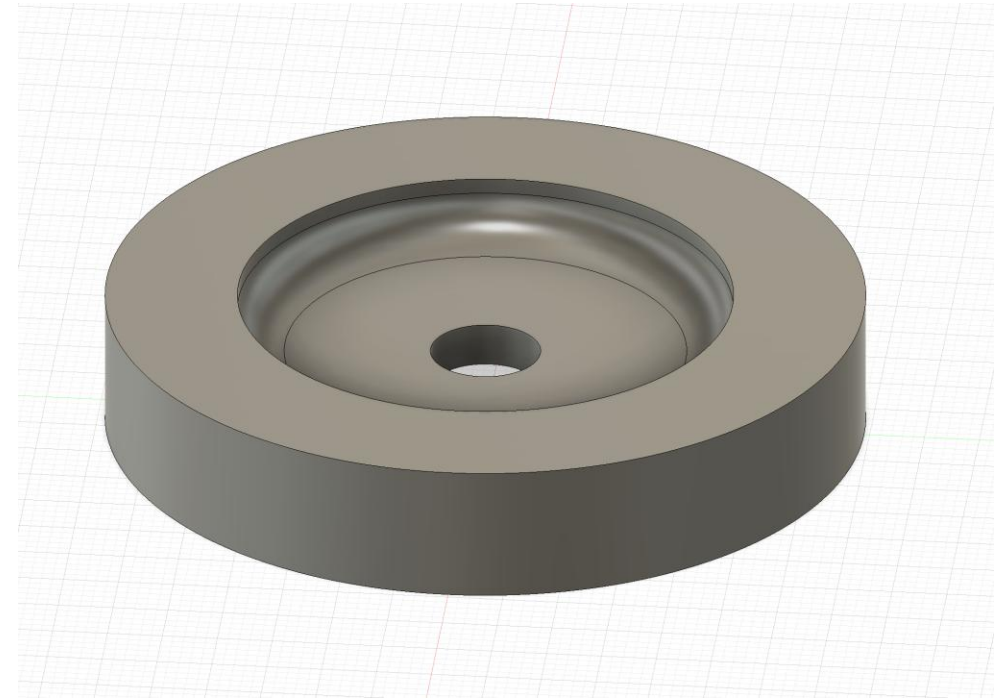
Female Plate

IJ

Manufacturing – CNC

Welded Plates

- Material: 1018 steel round stock
- CNC mill
 - 1st operation:
 - Work-holding: custom soft jaws
 - Face off top of stock, reduce diameter to 6.9"
 - 1" Center hole
 - 2nd operation:
 - Work-holding: custom soft jaws
 - Reduce thickness to 1.25"
 - Machine pocket and radius
- For one plate:
 - Waterjet tabs, drill press mounting holes
 - 3D print welding jig to ensure correct tab positioning on plate



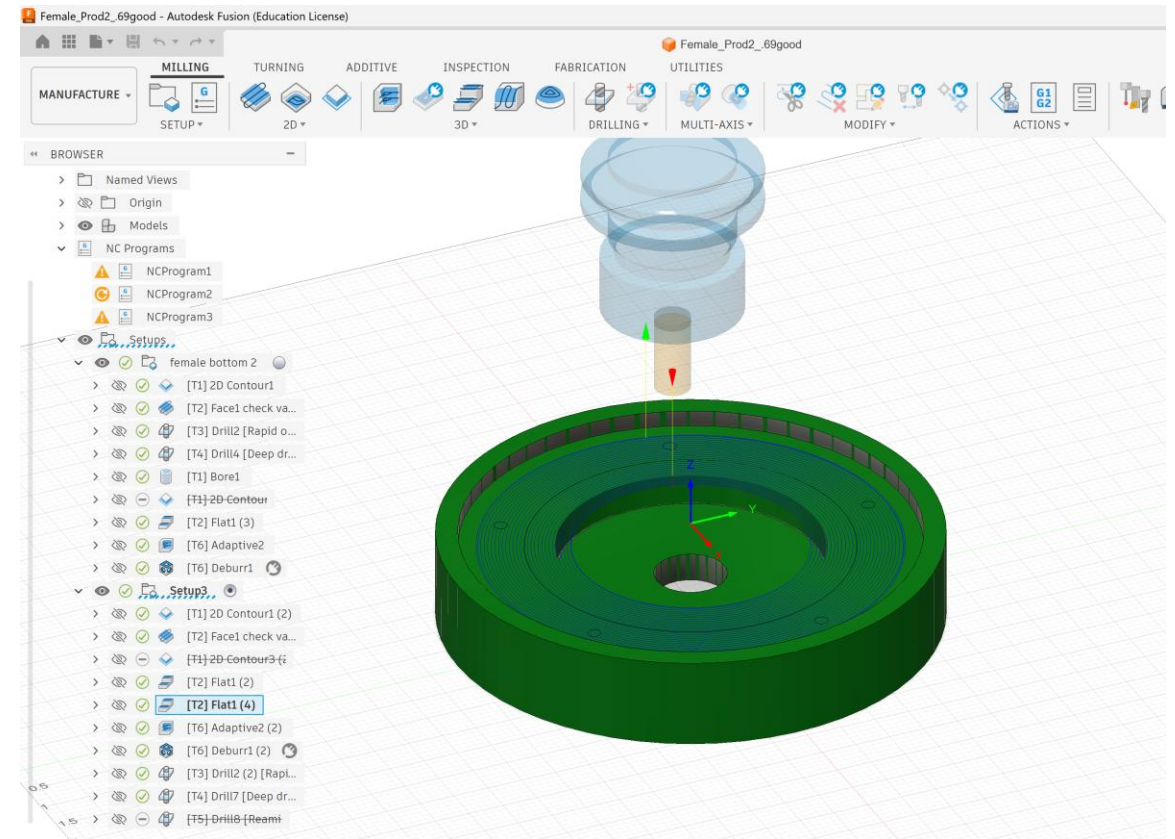
Welded plate

Manufacturing - Prototype

- Given the complexity of our project, we decided to create a prototype male plate to test and prove our manufacturing process before proceeding with our production models

Recap:

This was a wise decision because it taught use the tolerances possible on the TM-1 as well as prove out the CAM model, which allowed us time to practice and refine the milling process, reducing operating time from 8 hrs. to 5.5 hrs.



Manufacturing - Process



Step: 1
Home
machine to
stock center



Step: 2
Run
Operation 1.



Step: 3
Flip part,
recenter, and
run Operation 2



Step: 4
Remove
stock from
machine



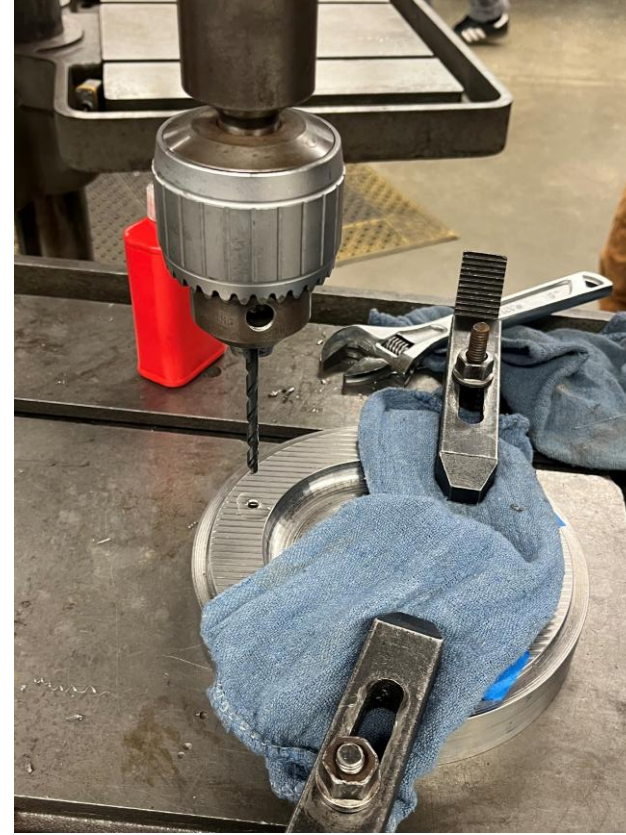
Step: 5
Clean
Finished
part

JV

Manufacturing - Process



Visual inspection



Upsize holes and ream

JV

Manufacturing - CNC Final



Welded Config



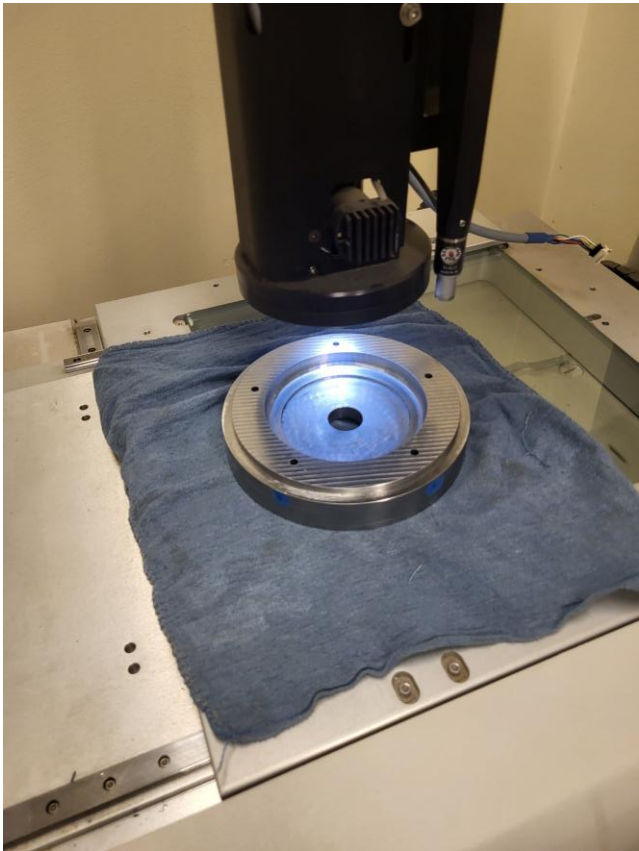
Male Config



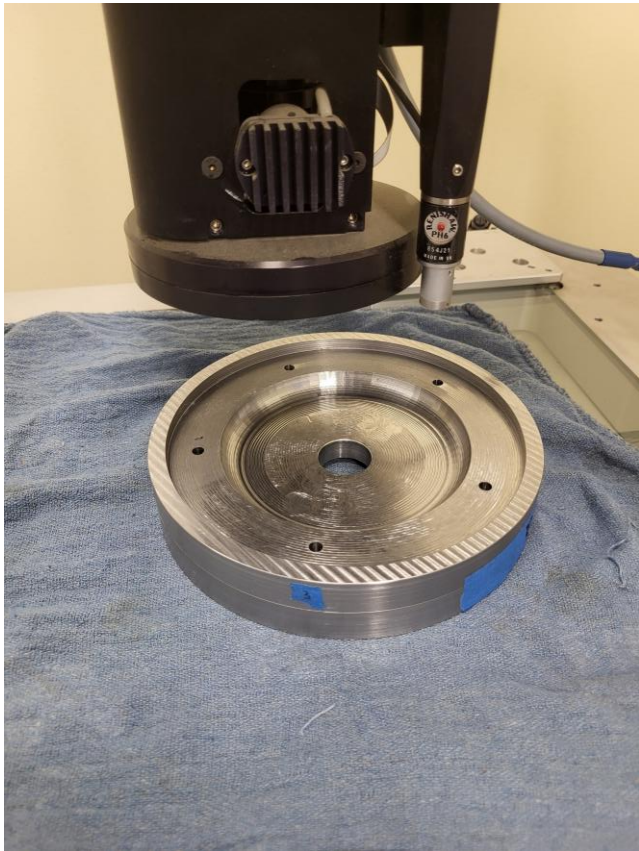
Female Config

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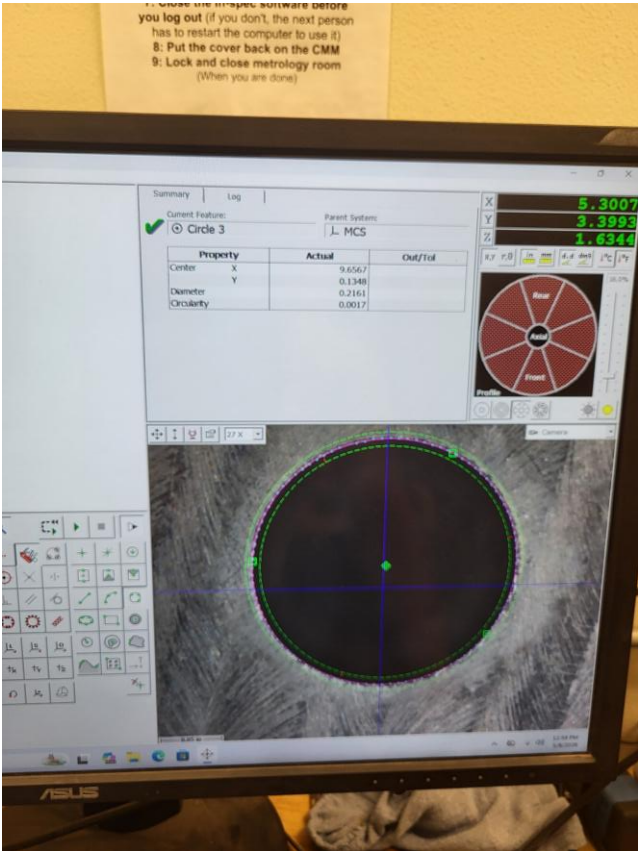
Measurements - CMM



CMM to measure hole diameters of male plate



CMM to measure hole diameter of female plate



diameter and circularity measurement

AP

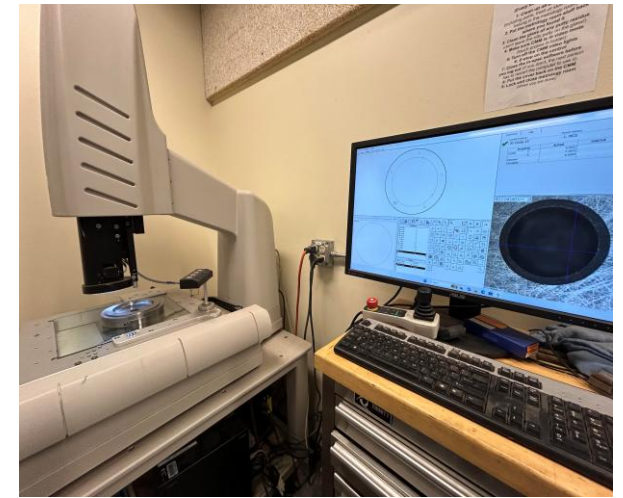
Manufacturing - Assembly

Critical Dimensions to Check on the CMM:

- Dowel pin hole diameter
- Dowel pin hole depth
- Pilot fit ledge depths and lengths
- Diameters of pilot ledges

Assembly with Interference Fits:

- Male plate in MATE department's oven: 200 degrees Celsius for ~75 minutes
- Dowel pins in freezer: ~75 minutes
- Press the dowel pins into the hot male plate (arbor press)
- Female plate in MATE department's oven: 200 degrees Celsius for ~75 minutes
- Male plate with dowel pins in freezer: ~75 minutes
- Press the male plate with the dowel pins into the hot female plate (hydraulic press)



JV

Manufacturing - Assembly



Heat up plates to 200°C while placing dowel pins in freezer



Press in dowel pins with arbor press



Prepare plates for final press



Hydraulic press both plates together

JV

Manufacturing - Welded Tabs



Step: 1
Clean and weld
on tabs



Step: 2
Visually inspect and
orient tabs to proper
location



Step: 3
Check tabs for
warping and
flatness

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Fixed-Free Testing



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Fixed-Free Modal Impact Testing

Initial testing method with the goal of validating ANSYS model of three configurations:

1. Mount plates to shake table with 3/8"-16 bolts
2. Attach accelerometer at the center of the top plate
3. Strike with impact hammer at 3 different places
4. Record data (3-point average)
5. Move attached accelerometer to a different position
6. Repeat steps 3-5 until last accelerometer position
7. Switch to next assembly, repeat steps 1-6



Accelerometer



Modal Impact Hammer

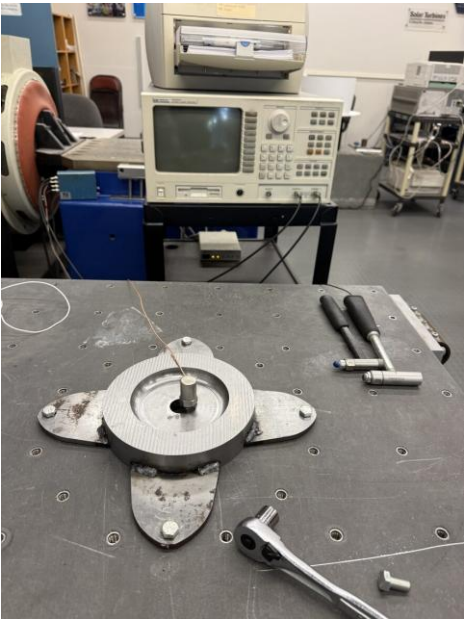
AP

Fixed-Free Testing Results

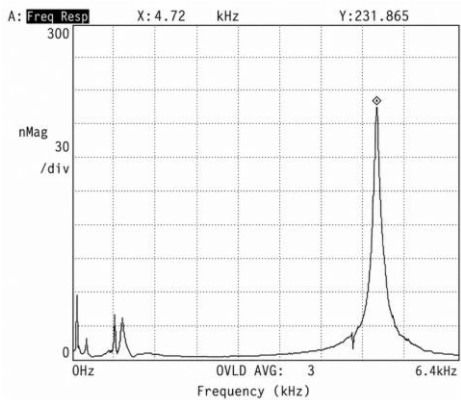
<u>Natural Frequency Comparison</u>	<i>Configuration 1</i>	<i>Configuration 2</i>	<i>Welded Configuration</i>
ANSYS	3.198 kHz	3.135 kHz	7.52 kHz
Experimental	4.72 kHz	4.912 kHz	7.06 kHz

Problems:

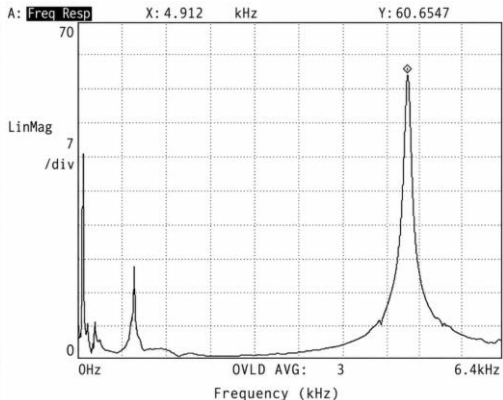
- Interference fit configurations
- Only looked at first mode shape
- Non-bolted tested gave inconsistent trends
- Critical contact surfaces were left out of ANSYS model



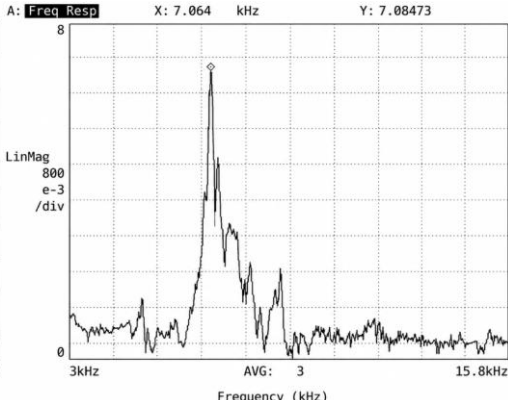
Testing area for Fixed-Free testing



Configuration 1



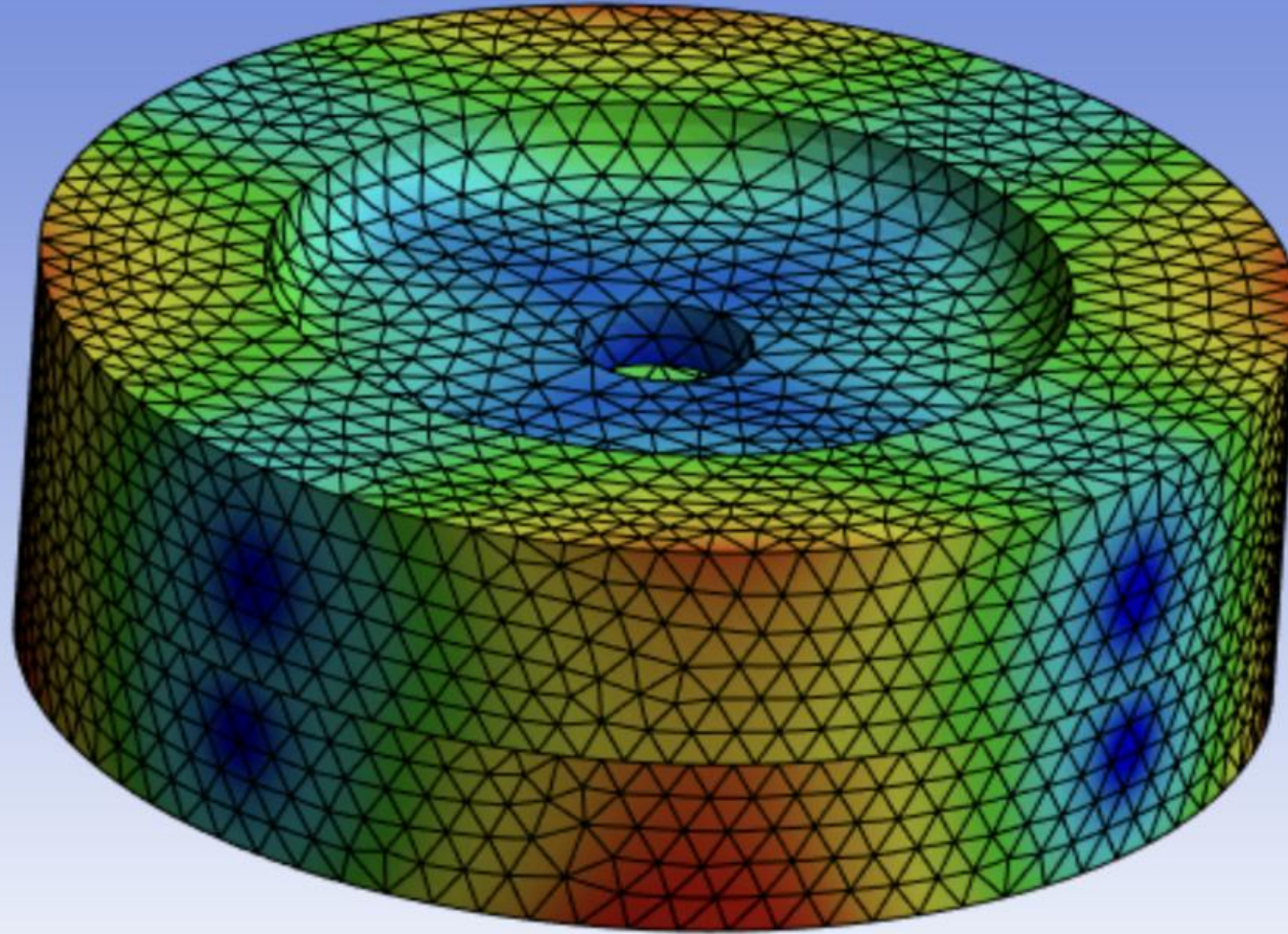
Configuration 2



Welded

AP

Refined ANSYS Model

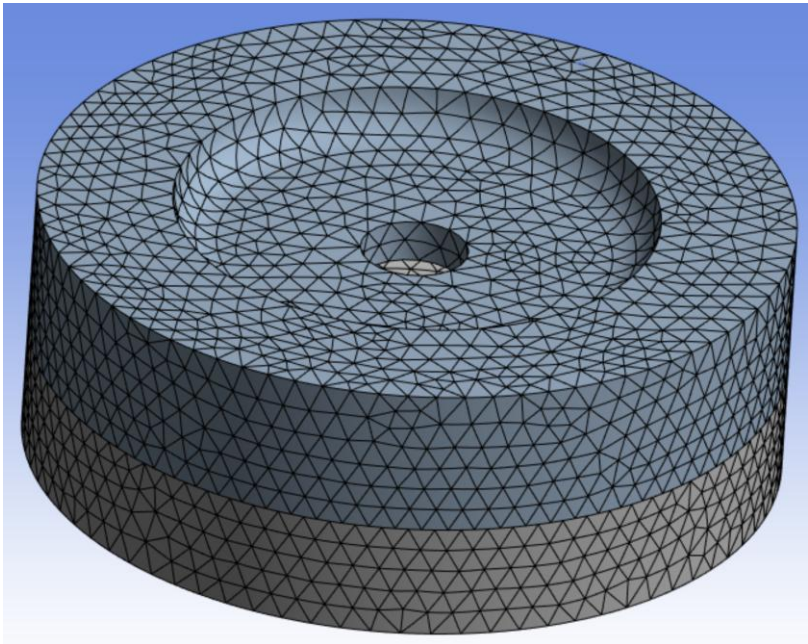


11

Refined Free-Free ANSYS Model

Improvements to ANSYS

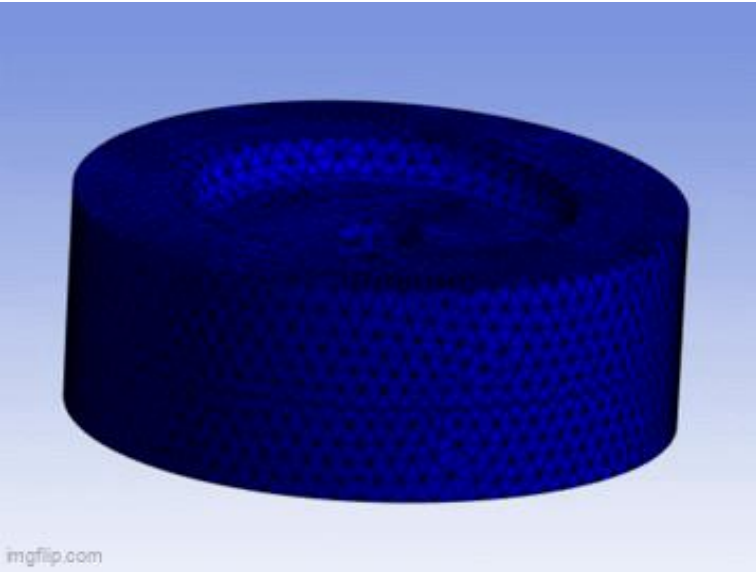
- Included contacts between plate surfaces
- Utilized weak springs to help convergence
- Removed fixed support
- Excluded tabs from ANSYS



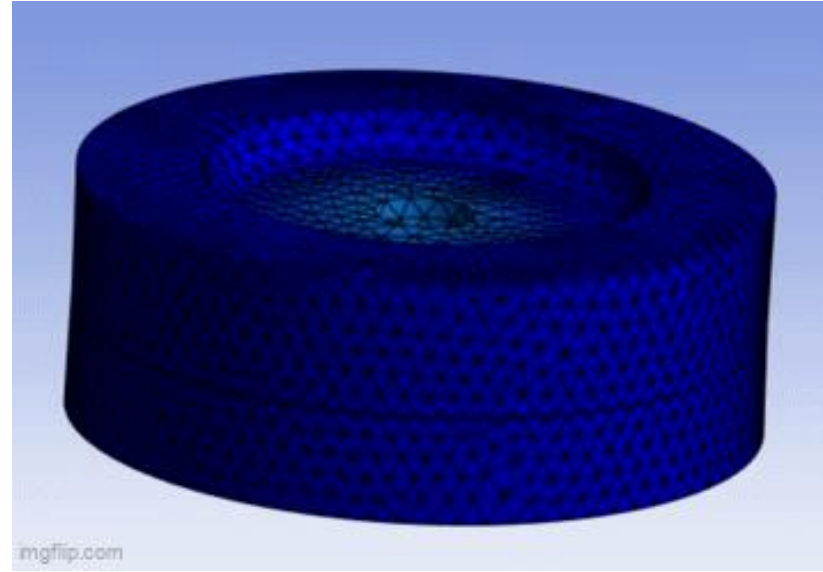
Configuration	Mode 6 (hz)	Mode 7 (hz)	Mode 8 (hz)	Mode 9 (hz)
1	0.196267677	4275.014956	4397.175	5655.72032
2	0.236836986	4268.660265	4826.458	5765.51508
Welded	7.55E-04	5891.8	7395.6	7592

IJ

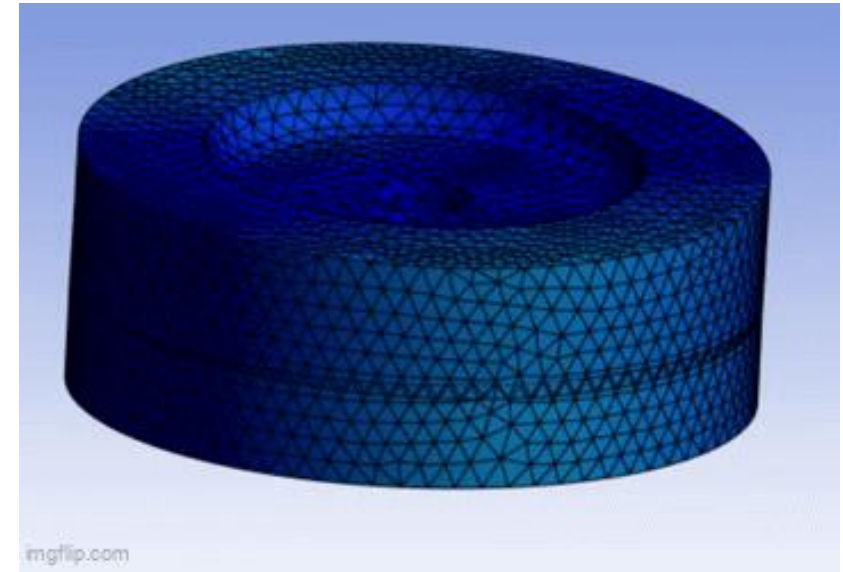
Mode Shapes for Interference Configurations



First Mode Shape



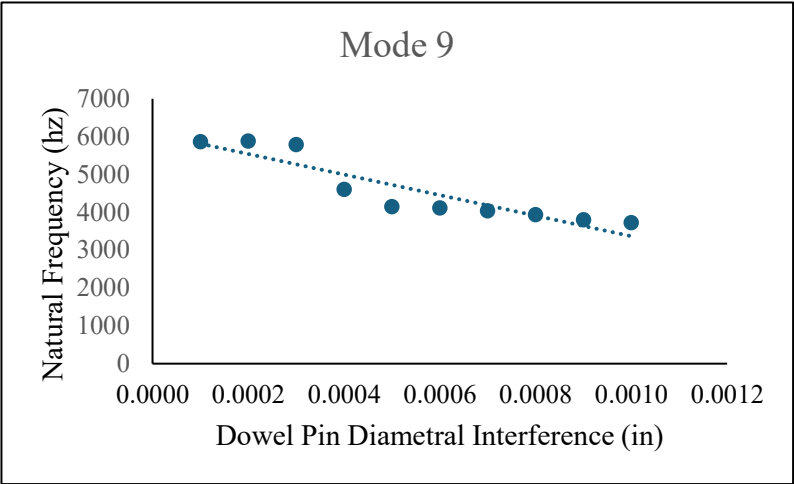
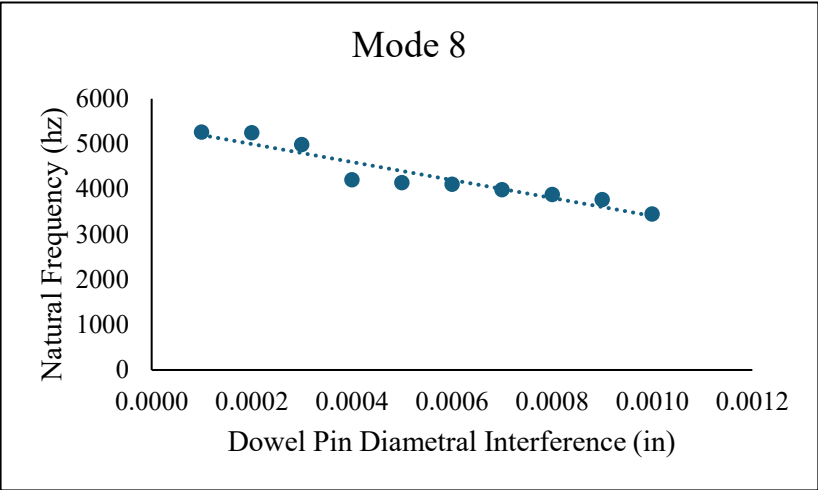
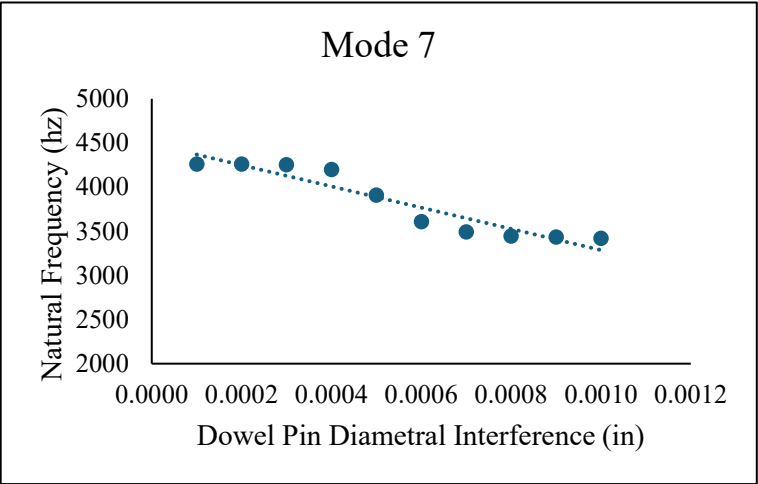
Second Mode Shape



Third Mode Shape

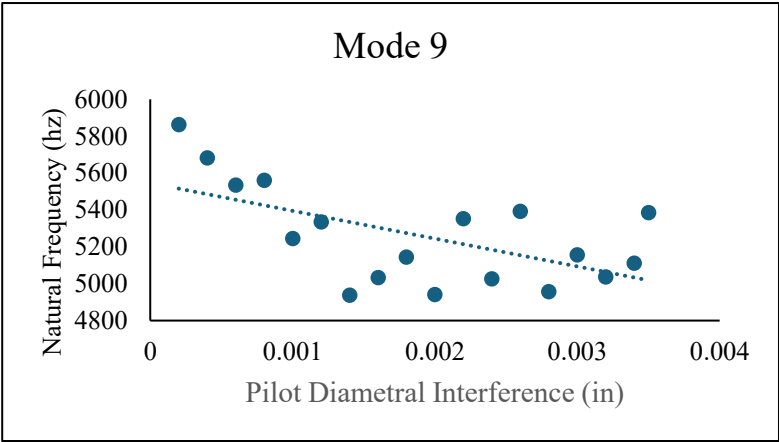
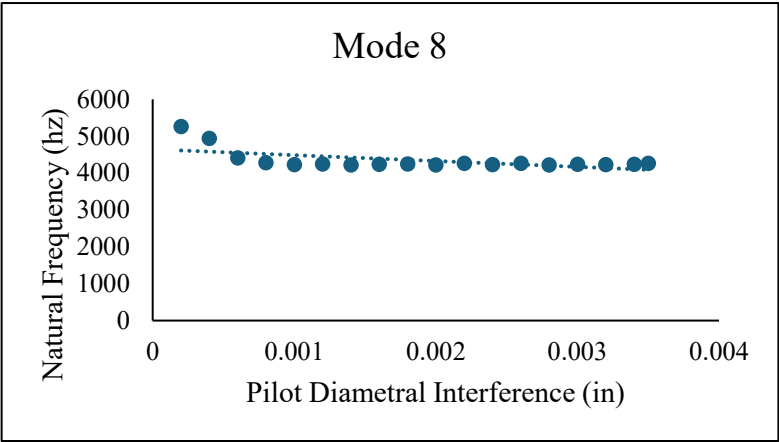
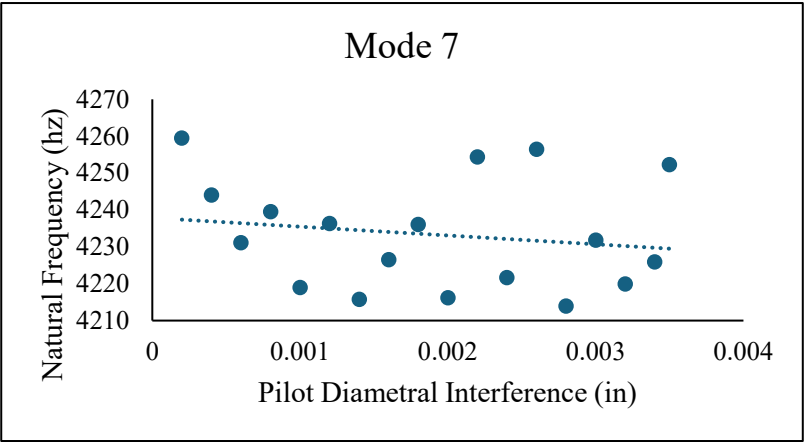
Refined Free-Free ANSYS Model Iterations

Dowel Pin Interference Iterations



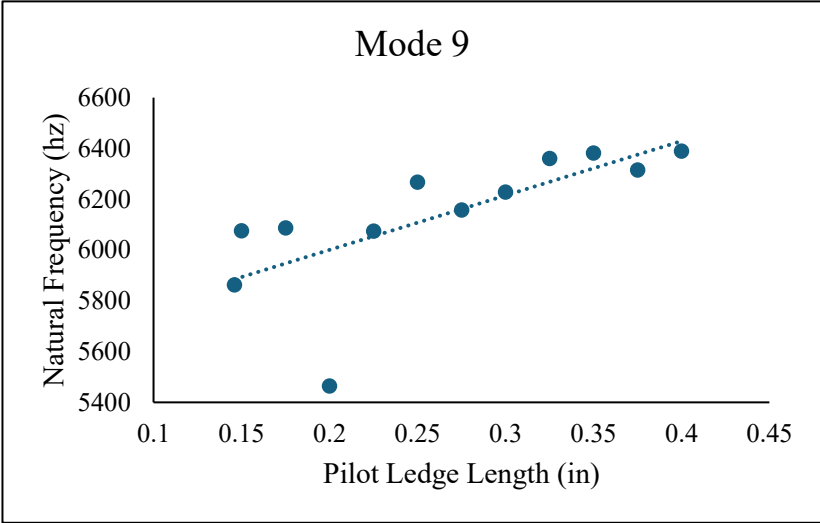
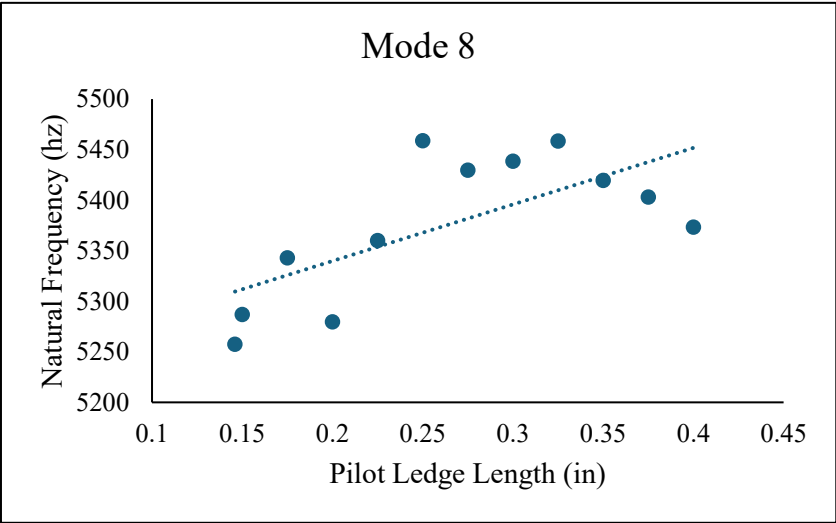
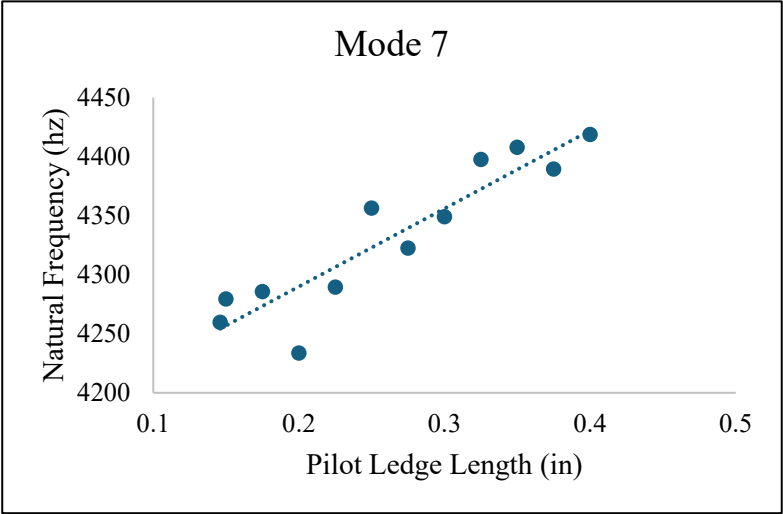
Refined Free-Free ANSYS Model Iterations

Pilot Interference Iterations



Refined Free-Free ANSYS Model Iterations

Pilot Ledge Contact Length Iteration



Free-Free Testing

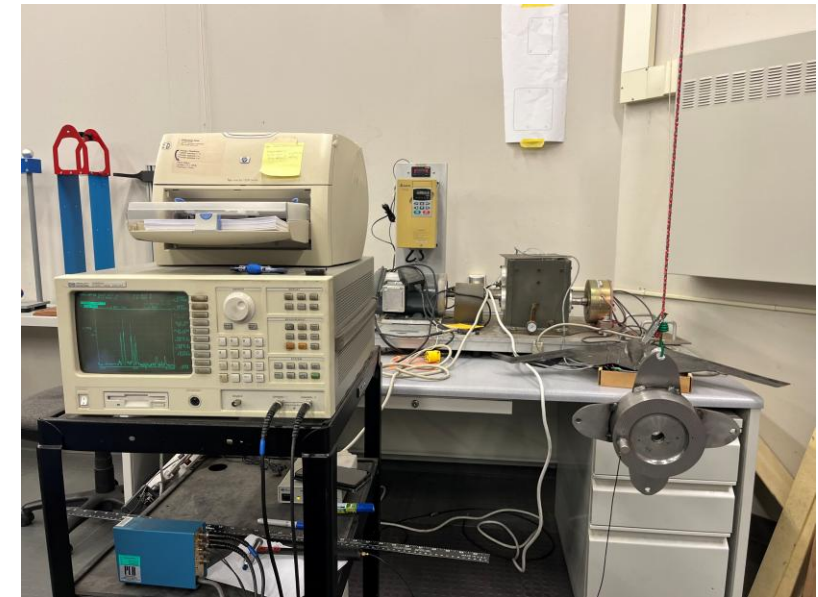
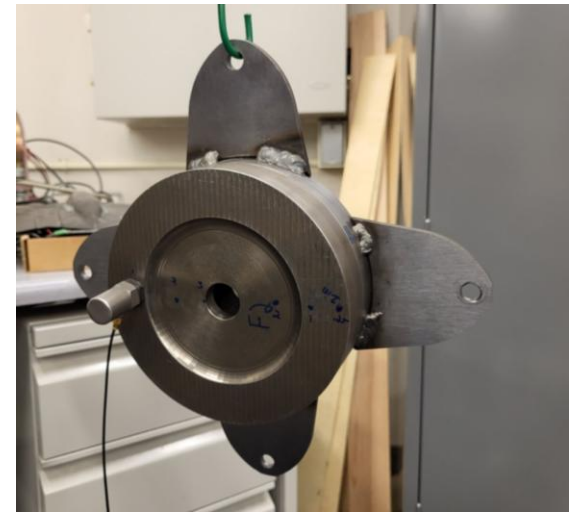


JV

Free-Free Modal Impact Testing

Refined testing method with the goal of validating new ANSYS model for the three configurations:

1. Hang plate assembly in a free-free condition
2. Attach accelerometer at the center of the top plate
3. Perform roving hammer test
4. Record data (3-point average)
5. Perform roving accelerometer test
6. Record data (3-point average)
7. Switch to next assembly, repeat steps 1-6



JV

Free-Free Testing Results

<u>Second Natural Frequency Comparison</u>	<i>Configuration 1</i>	<i>Configuration 2</i>	<i>Welded Configuration</i>
ANSYS	4.397 kHz	4.826 kHz	5.892 kHz
Experimental	4.384 kHz	4.608 kHz	4.88 kHz

<u>Third Natural Frequency Comparison</u>	<i>Configuration 1</i>	<i>Configuration 2</i>	<i>Welded Configuration</i>
ANSYS	5.655 kHz	5.765 kHz	7.592 kHz
Experimental	5.392 kHz	5.935 kHz	8.033 kHz



Noticeable Differences:

- Bode plot data appeared cleaner
- Experimental results were closer to the ANSYS model
- Were able to see second and third mode shapes

AP

Cost Breakdown – Assembly Costs

Assy Level	Part Number	Descriptive Part Name				Qty	Part Cost	Total Cost	Source	URL	More Info
		Lvl0	Lvl1	Lvl2	Lvl3	Lvl4					
0		Final Assy									-
1	1000	Assembly 1									-
2	1100			Male Plate			1	\$ 55.80	\$ 55.80	Custom	-
3	1110				Mounting Tabs		4	\$ 5.00	\$ 20.00	Custom	-
4	1111					3/8"-16 Bolt	4	\$ 0.55	\$ 2.20	McMaster https://wv	-
2	1200			Female Plate			1	\$ 55.80	\$ 55.80	Custom	-
2	1300			7/32" Dowel Pins			5	\$ 0.21	\$ 1.05	McMaster https://w	-
1	2000	Assembly 2									-
2	2100			Male Plate			1	\$ 55.80	\$ 55.80	Custom	-
3	2110				Mounting Tabs		4	\$ 5.00	\$ 20.00	Custom	-
4	2111					3/8"-16 Bolt	4	\$ 0.55	\$ 2.20	McMaster https://wv	-
2	2200			Female Plate			1	\$ 55.80	\$ 55.80	Custom	-
2	2300			7/32" Dowel Pins			5	\$ 0.21	\$ 1.05	McMaster www.mcm	-
1	3000	Assembly 3									-
2	3100			Male Plate			1	\$ 55.80	\$ 55.80	Custom	-
3	3110				Mounting Tabs		4	\$ 5.00	\$ 20.00	Custom	-
4	3111					3/8"-16 Bolt	4	\$ 0.55	\$ 2.20	McMaster https://wv	-
2	3200			Female Plate			1	\$ 55.80	\$ 55.80	Custom	-
Total Parts						40		\$ 403.50			-

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Cost Breakdown – Tool Costs

Tooling Costs

Tool	Qty	Part Cost	Total Cost	Source
Endmills	6	\$52-\$112	\$ 463.11	McMaster
Reamers	6	\$25-\$30	\$ 191.37	McMaster
Vise Jaws	4	\$58-\$64	\$ 239.98	McMaster
Stock Plates	9	\$74.87	\$ 673.83	McMaster
Drill Bits	2	\$23.00	\$ 30.93	McMaster
Total Parts		15	\$1,599.22	

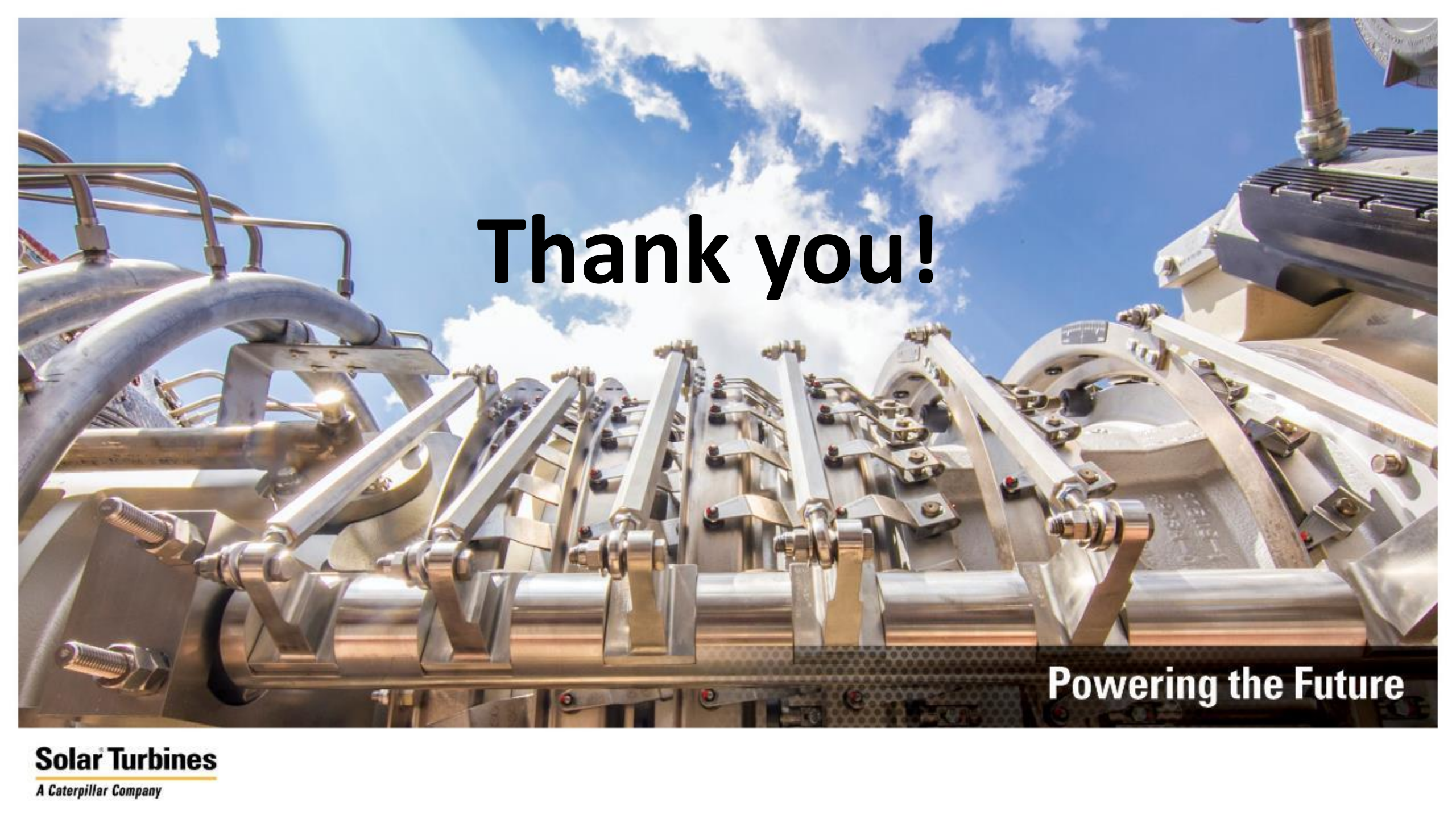
Miscellaneous Costs: \$133.95

Total Spent: \$2,136.67

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Conclusions

- Increasing pilot ledge length was the main contributor to the increase in natural frequency and subsequently an increase in overall assembly stiffness
- Configuration 2 consistently produced higher frequencies than Configuration 1 in both ANSYS and experimental testing.
- The welded configuration produced the highest stiffness and natural frequencies overall.
- ANSYS predictions closely matched experimental trends, validating the simulation approach.
- Differences between simulation and testing were likely caused by manufacturing tolerances, accessibility to good tooling, and testing inconsistencies.



Thank you!

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